



Youth Destress



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Background of Project Idea

- Machine Learning and Neural Networks
- Project Idea Title 1: Deep Learning on a public dataset
- Mental Health has been a pressing issue in many countries around the world.
- A 2012 report by the American Academy of Pediatrics states that Social media affects youths more than adults might know of (*Nothaft, 2023*).
- It also causes anxiety and sleep deprivation which affects the growth of youths.
- In this digital world with the high prevalence of social media platforms and interactions, it is unlikely for youths and young adults to stay away from using social media platforms.



Background of Project Idea

- They have become a part of our lives and act as a form of de stressing activity. How is the use of social media related to mental health in youths and young adults?
- According to an interview by TODAY newspaper, teenagers and young adults stated that social media addiction has negatively affected their lives and emotions(Lau, 2023).
- Hence, it is important to address this issue and provide more concrete evidence stating that high social media usage affects the mental health of youths and young adults.

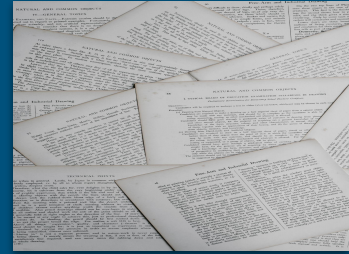


PROBLEM STATEMENT



- The Aim is to build a model that is able to predict whether high social media usage affects mental health in youths and young adults.
- The target audience are Youths and Young adults.
- Youth Destress model will contain features that are able to accurately identify youths and young adults who are at risk.
- Insights from the research papers I have analysed will be incorporated into the Youth Destress model.

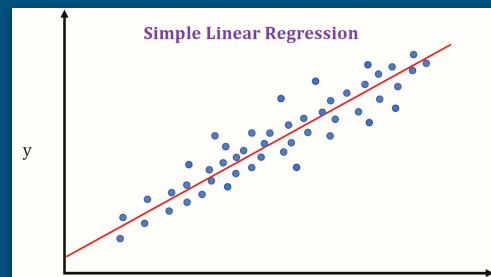
RESEARCH METHODOLOGY



- Multivariate regression models are able to successfully identify how prolonged use of social media causes depressive symptoms among youths based on average number of hours spend on social media(*Kelly et al., 2019*).
- Support Vector Machine(SVM) and Logistic Regression were able to provide the most precise results when applied to four different Twitter datasets related to depression and sentiment analysis from Kaggle(*Obagbuwa et al., 2023*).
- Logistic Regression performed the best at analysing live tweets for any signs of depression as it had the least execution time and more accurate results than SVM(*Obagbuwa et al., 2023*).

RESEARCH METHODOLOGY

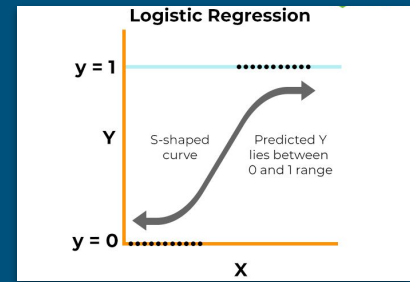
- Multilevel linear regression model was unable to predict whether social media use at ages 12-14 affected the mental health at ages 14-15 subsequently(*Plackett et al., 2022*).
- Other factors were at play such as self-esteem(*Plackett et al., 2022*).
- Simple linear regression model showcases a clear link between social media addiction and depression among University Students(*Haand & Shuwang, 2020*).



TECHNIQUES TO TRAIN THE MODEL

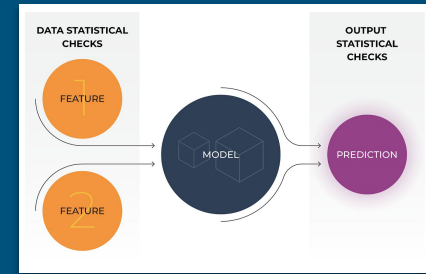
- The model can make use of logistic regression model as a framework to analyse the dataset and classify the data accordingly.
- It is important to take into consideration other factors that might affect the output of the model such as self-esteem, sleep etc.
- Other machine learning models must also be used to compare the performance of the chosen model.
- Model Accuracy should be calculated to evaluate the performance of the model.
- Outliers and Missing Data must be treated accordingly to prevent inaccuracy in results.

YOUTH DESTRESS



- Youth Destress will use a logistic regression model to analyse and identify youths who are at risk of developing mental health problems.
- The dataset that will be used to train the model is a dataset related to social media and depression among youths obtained from Kaggle.
- The model will classify youths into three groups according to the hours they spend on social media sites.
- If they spend more than 5 hrs, then they will be classified as serious and at risk. 3-5 hrs is moderate and less than 2 hours is considered as safe.

OUTPUT OF THE MODEL



- The results will be presented in a graphical form showcasing the probabilities of the three groups of classification.
- The results will display the participant as either serious, moderate or at risk after their data is imputed into the model.
- The Youth Destress model's accuracy will be calculated using confusion matrix and its accuracy score will be used to evaluate the performance of the model.
- Other models such as SVM and XGB Classifier will also be used as baseline models to compare the performance of Youth Destress model.

CONCLUSION

- The Youth Destress model is modelled after the insights and techniques analysed from research papers.
- The model is designed to identify youths at risk of developing mental health issues and the results will be presented in a graphical format and classified into a group accordingly.
- The model's performance and accuracy will be evaluated accordingly to ensure that the output is accurate and the model is efficient.
- Other models will also be used to compare and evaluate the performance of the model.
- Further market research will be performed to improve the model's features.

Thank You

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